

Production of Noble Metal Powders

Oleg D. Neikov*, Stanislav S. Naboychenko[†] and Irina B. Murashova[†]

*Frantsevich Institute for Problems of Materials Science (IPMS), Kiev, Ukraine, [†]Ural Federal University (UrFU), Yekaterinburg, Russia

Chapter Outline

Introduction	831	Powders of Platinum Metals	837
Production of Silver Powders	832	Chemical Reduction	839
Chemical Processes	832	The Carbonyl Process	840
Atomization	833	Autoclave Precipitation	841
Mechanical Comminution Processes	833	Production of Noble Metal Alloys Powders	842
Production of Silver Powder From Solution by Electrolysis	834	Production of Platinum Black Deposit (Niello)	844
Production of Silver Powder by Chemical Reduction With		Synthesis of Palladium Nanopowders by Methods	
Electrochemical Regeneration of the Reducer	835	of Zone Electrochemistry	845
Production of Fibrous Dendritic Silver Residue From		Producing of Noble Nanoparticles in Metal-Organic	
Molten Electrolytes	835	Frameworks Pores	845
Production of Gold Powders	835	Noble Metals Application in Implantable Medical Devices	845
Production of Platinum-Group Powders	837	References	846

INTRODUCTION

The properties of noble metals including silver, gold, and platinum-group metals are listed in Table 25.1.

Chemical precipitation, thermolysis, electrolysis, hydrogen reduction, and melt atomization are often used for the production of powders of these metals. The basic materials are solutions, salt pulps, and hydroxides; in the latter case, solid-phase properties significantly influence the properties of the powder produced, complicating the reproducibility of a number of its characteristics (particle size, bulk mass, surface).

Among organic reagents, such as formaldehydes, alcohols, oxalic and formic acids and their salts, and glycerin, and among nonorganic ones, such as hydrazine and its derivatives, borates of alkali and alkaline-earth metals, hypophosphites, Ti (III), Cr (III), and V (III) salts, are used most often. Organic reducers are safe, available, and allow carrying out the process at low temperatures. The produced powders have a developed surface; they are also dispersed (1.0–10 μm), but are gas-saturated and require a special finishing procedure. Formic acid and its salts, sodium formate in particular, are used most often. This is an available, nontoxic, and easily soluble reagent. The precipitation factors depend on temperature, pH value, and reagent consumption, but there is a problem of washing the powder free from sodium ions resulting in high water consumption (up to 160–200 L per 1 kg of precipitate). The particle size of the produced residue is between 0.05 and 0.5 μm.

Reney's method is based on producing the alloy, followed by selective leaching of the less noble component by acid or alkali solutions. The produced precipitates of metals or their alloys have a large specific surface (80–100 m²/g), but the residues of the leached metal contaminate them, so they need to be passivated.

Thermolysis means the thermal decomposition of metal salts (nitrates, halides, ammoniates, carbonates, formates), which form volatile products upon heating. The powders produced are distinguished by a nonuniform granulometric composition, a small specific surface, and contamination by thermolysis products.

Electrochemical methods (extraction from solutions, processing of salt pulp and hydroxides, electrolysis of wastes and recycled raw materials) are widely used in the production of noble metal powders. Powder properties depend on the electrolyte content, the current density, the hydrodynamic conditions, and the cathode shape.

A solid-phase reduction of salts and oxides of noble metals by hydrogen at 500–700 K is possible; however, the powders produced (especially palladium powders) are inclined hydrogenation and contain up to 0.1% of adsorbed hydrogen. Indices

TABLE 25.1 Basic Properties of Noble Metals

Characteristic	Metals							
	Ag	Au	Ru	Rh	Pd	Os	Ir	Pt
Serial number	47	79	44	45	46	76	77	78
Atomic mass, g/mole	107.9	197	101.1	102.9	106.4	190.2	192.2	195.1
Relevant oxidation rates			3, 4, 6–8	3	2	4,6–8	3–4	2,4
Density, g/cm ³	10.49	19.32	12.45	12.41	12.02	22.61	22.65	21.45
Temperature, K:								
of melting	1234	1337	2856	2239	2098	3306	2739	2041
of boiling	2485	3080	4423	4000	4516	5285	4701	4098
Specific heat capacity, J/(K·mole)	25.4	25.2	24.0	25.1	26.0	24.8	25.1	25.9
Thermal conductivity, W/(K·m), at 298 K	433	318	117	152	75.2	87.6	147	74.6
Specific electrical resistivity, nΩ·m, at 293 K	16.1	22.1	71	43.3	105.1	81.2	47.1	105

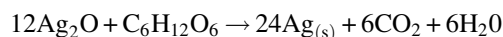
of hydrogen reduction (purity and physical properties of the powder) are significantly improved when the hydrothermal method is used, by precipitation of metals from an organic aqueous medium at elevated temperature (380–450 K).

PRODUCTION OF SILVER POWDERS

Basic methods of production are solid-phase reduction, chemical precipitation from solutions or pulp, electrolytic precipitation, and electrolysis.

Chemical Processes

Most silver salts are easily reduced from the solution, with the only limitation being their solubility. The most common compounds are silver nitrate and silver oxide. Organic reducing agents such as alcohol, sugar, or aldehydes are used to manufacture silver powder from silver oxide. For example, dextrose can be used to produce a very fine silver powder:



Organically precipitated silver powders have particle sizes ranging from 5 μm to less than 0.1 μm. However, these silver powders tend to agglomerate into aggregates 10 μm in size or larger. To avoid agglomeration, the powder can be precipitated in the presence of a surfactant or deagglomerated after the precipitation using a mechanical technique.

Silver powders are produced by mixing an acidified solution with a solution of tin (II) salts in the presence of anion-type surfactants, controlled pH values, and temperature.

To produce submicron particles of silver and palladium, their concentrated nitrate solutions are used, aerosol forming and drop separation by size are provided, and then drying and conversion of aerosol particles to the metallic condition are performed in a reducing atmosphere at elevated temperature.

Particles 7–10 μm in size are produced by the evaporation of silver at 1270 K with subsequent condensation in a nitrogen atmosphere.

Metal-ceramic contacts of heightened wear resistance are produced from silver-based compositions (Ag–oxides of cadmium, copper, zinc, tin, or lead, and Ag–Ni) from nitric-acid solutions of appropriate composition; the metals are jointly precipitated in the form of hydrates and carbonates. Powder mixtures with a particle size of 0.1–0.4 μm are produced after separation of precipitates, followed by their washing, drying, and calcination at 670–770 K. To provide a heightened flow

rate, an agglomeration is carried out; the size of the produced particles is 200–400 μm . Removal of cadmium oxide proceeds during sintering of products containing cadmium oxide at $T > 970\text{ K}$; a contact film of silver, which excludes oxidation, is formed in the surface layer.

A composition for the products that retains their electrical conductivity and wear resistance at low temperatures (for example, electrical contacts) and in a vacuum is produced by mechanical stirring of dispersed silver powders with additives of 2%–49% of rhenium or molybdenum disulfide in water or alcohol.

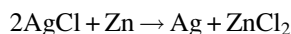
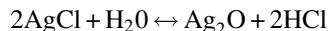
Silver powder is produced by processing a solution of silver nitrate (0.05 M) and 4-(methylamino)phenol sulfate in the ratio of $[(\text{C}_7\text{H}_9\text{ON})] \cdot \text{H}_2\text{SO}_4/\text{AgNO}_3 = 1.05$. The precipitate is washed by a 3% solution of sodium nitrite, by hot water (330–340 K), and dried at $373 \pm 5\text{ K}$; the powder contains not less than 98.7% of Ag. The particle shape is regular polyhedrons 0.5–20 μm in size or ovals 0.3–4.5 μm in size. The powder surface is 0.19–0.21 m^2/g , the apparent density is 2.4 g/cm^3 . The powder is used for film covering production (70% of Ag) with specific electric resistance not higher than $1.2 \times 10^{-7} \Omega \cdot \text{m}$.

Silver powder is also produced by processing hydroxide pulp with a solution of sodium formate; the precipitate is washed by water and dried. Powder properties (specific surface, S , apparent density, ρ , median mass diameter, d_m) significantly depend on the reduction conditions:

	S (m^2/g)	ρ (g/cm^3)	d_m (μm)
Production conditions			
without surfactants	4.9	0.98	Aggregates
with surfactants	0.45–1.25	1.6–2.6	0.5–2.5
$L:S = 10$ (relation of liquid to solid)	1.3	...	10
$L:S = 1$	0.8	2.9	17

The consumption of the reducing agent is 1.1%–2.4% of the precipitated powder mass with a content of 99.9% of Ag.

It is possible to produce silver powder by a self-propagating low temperature reaction of precipitated silver chloride with zinc. In this process, powder of silver or a “silver-zinc” alloy of the Ag_nZn_m type is produced by means of chemical reactions considered as the kind of contact displacement:



$$\Delta G = -150\text{ kJ/mole}$$

Zinc consumption is stoichiometric; the duration of the process is 15–20 min, the transformation level is close to 100%. Powder surface area is 0.8 m^2/g , the particle size is 1–2 μm , and their shape is round.

Atomization

Silver powder is produced by the atomization of a molten silver stream with gas or liquid. Average particle sizes generally are within the range of 10–50 μm . The gas-atomized powder generally approaches a spherical shape with a smooth surface, minimal porosity, and a high powder density [1].

The spray pyrolysis technique for the manufacture of silver powders is also used. This process involves spraying a silver nitrate solution into a flame formed by a coaxial burner. Control of the flame temperature and spray conditions allows for the control of particle size and shape [2].

Spray pyrolysis produces solid spherical dense silver powders. An aerosol of silver nitrate solution is heated to temperatures as low as 873 K in nitrogen to decompose the silver salt and to produce a fully dense silver powder. The particle size of the powder is 5 μm and smaller.

Mechanical Comminution Processes

Mechanical milling is used to change the properties of silver powders, including disintegration of particle aggregates, particle shape, and particle surface characteristics. This process can be used for most types of silver powders with techniques such as ball milling, vibratory milling, or attrition milling.

A feature of these processes is that they are time consuming and require special precautions to prevent contamination from the mill and milling medium. Various types of milling media can be used, including zirconia balls (ZrO_2), metallic balls, and glass balls. Continued milling produces two-dimensional silver flakes. The sizes of the milling media is between 10 and less than 1 mm, depending on the sizes of the initial powder and the size of the resulting silver flake required. Generally, an organic surfactant is introduced during the milling process to prevent the agglomeration of particles with each other while grinding and to control the size and shape of the silver flake.

Production of Silver Powder From Solution by Electrolysis

The basic raw material is Dore's alloy that contains: 87%–95% of Ag; 2%–14% of platinoids and gold; 1%–4% of Cu; 0.06%–0.01% of Ni; 0.005%–0.01% of As; 0.01%–0.05% of Sb; 0.1%–0.2% of Fe; 0.001%–0.005% of Pb; 0.002%–0.01% of Se; 0.005%–0.1% of Te; and silver-containing materials (80%–84% of Ag; 16%–20% of Cu). In accordance with State Standard 28595-90, silver of three brands (CpA-1, CpA-2, and CpA-3) is used in the cells with soluble anodes (respectively 99.99%, 99.98%, and 99.9% of Ag).

Silver is precipitated on the cathode as a loose needle-like deposit that is easily stripped from the cathode. The current efficiency in the nitrate solution is approximately 100%. The cause of electric loss is short circuits of silver dendrites and simultaneous reduction of NO_3^- anions to HNO_2 , NO_2^- , NO_2 (gas), or NO (gas).

The respective O/R potentials are 0.94, 1.01, 0.77, and 0.95 V (nhe). Therefore the concentration of the free acid is decreased.

Fine silver powder is precipitated at the silver anode due to the $\text{Ag}_2^+ = \text{Ag} + \text{Ag}^+$ reaction. Practically all impurities except gold and platinoids pass into the solution. Part of the palladium can pass into the solution from platinoids. Selenium and tellurium should be eliminated from the anode material during fire refining.

The powder is produced in cells with vertical or horizontal cathodes (Table 25.2) as well as on a rotating cylindrical cathode. Vertical cathodes in cells are sheets of silver, high-clean aluminum, or stainless steel. The intercathode space is 16–20 cm. The anodes are put into a diaphragm box with stretched bags from polyvinyl chloride (PVC) to collect the anode sludge. The ends of the cathode rods are pushed into slots welded to a bus. Vertical plates that perform reciprocal movements are installed for deposit removal. The silver deposit is stripped into a box with a filter bottom. A telfer for lifting the frame and the box with silver powder is installed above cells. The powder is removed by knives from the cylindrical rotating cathode, and a bottom knife transfers the deposit to the sloping conveyor.

Cells with a horizontal location of electrodes are manufactured of plastic or ceramics and lined with acid-proof tiles. A graphite dalle (cathode) is embedded into the cell bottom. Half the cell bottom length is splayed to provide powder deposit discharge. An anode box with the bottom made of fire bars, on which a filtration nylon material, plastic grid, and some layers of anode plates are placed, is installed above the horizontal part of the cell. The current is supplied by heavy contacting leads, to which the ends of a flexible cable are soldered. Anodic contacts are manufactured of 50% Ag+50% Au alloy; cathodic contacts are fabricated from silver. Once every day, the deposit is discharged by knives on the filtration frame that is brought to the cell by a car. The deposit is washed by hot water acidified by nitric acid

TABLE 25.2 Electrolysis Conditions and Process Parameters

Parameters	Cell With Vertical Electrodes	Cell With Horizontal Electrodes
Electrolyte concentration, kg/m ³ :		
Silver	10–20	40–50
Nitric acid	10–15	15–25
Cathodic current density, A/m ²	1500	400–500
Temperature, K	313–343	313–323
Deposit growth time, min	2–10	1440
Cell voltage, V	2–2.8	3.2–3.8
Current efficiency, %	95–97	87–90

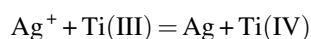
(1 kg/m³). Cleaning the solution includes the elimination of lead and copper. Lead is precipitated by sulfuric acid. To eliminate copper, at first silver cementation from the solution is performed by copper plates, then the deposit is removed, and copper cathodes and insoluble magnetite anodes are put into the cell; the copper concentration is decreased to 10 kg/m³ during electrolysis.

It is possible to produce silver powder in the cells with insoluble anodes. Titanium sheets are covered by a thin platinum layer. Anode current density is twice as high compared with the cathode one. The nitric acid concentration increases during electrolysis whereas the silver ion concentration decreases. Periodically, a part of the electrolyte is vented away for regeneration and removal of platinum metals and gold.

Silver powder produced by electrolysis meets the following requirements: >99.9% of Ag, not less than 0.02% of Cu; 0.04% of the sum (Fe, Pb, Al, Bi), 0.001% of NO₃⁻, 0.005% of SO₄²⁻, 0.08% of moisture (over 100%); particle size for PS1 brand: 97% of the size fraction of (–56) μm, for PS2 brand: and 96% of the size fraction of (–160) μm.

Production of Silver Powder by Chemical Reduction With Electrochemical Regeneration of the Reducer

Fine silver powder is produced in the anodic space of the electrolyzer close to the surface of soluble silver anodes by chemical reduction with titanium (III) ions:



The composition of the solution is 10% of Ti₂(SO₄)₃ and 10%–15% of H₂SO₄. The electrolysis is carried out at 298 K, providing cathode and anode current densities of 50–100 and 250 A/m², respectively. The fine powder formed with a particle size of 1 μm is precipitated on the bottom of the electrodes. The solution of titanium sulfate goes to the cathodic space where titanium (IV) ions are reduced to Ti (III) ions. The powder is washed on a filter by 5% solution of sulfuric acid to eliminate titanium and then by water to eliminate sulfate ions. The water residue is eliminated by alcohol or acetone washing. The final drying of the powder is performed under vacuum at 315–325 K.

The several ways of production of electrolytic silver and silver-alloy powders are shown in Table 25.3.

Production of Fibrous Dendritic Silver Residue From Molten Electrolytes

Electrolysis is carried out using a silver nitrate melt based on the potassium-sodium eutectic (15–33 wt% of silver) at a temperature of 523–563 K in an open cell without additional stirring [8]. A flat anode in the form of a silver bar with insulated silver current supply is placed on the cell bottom. The cathode is a silver wire ($d = 1.5$ mm) clamped at the top between rolls of the lifting device. Prior to turning on the main current, a current of 0.1–0.2 A is applied for 1–2 min to provide stronger adherence of the fibrous deposit to the cathode. Without current interruption, the fibrous deposit is precipitated on the sublayer, simultaneously providing cathode lifting (v) and current (i_a) regulation to the required level (in the range of 0.5–10 A). The speed of cathode lifting (from 0.25 to 12 cm/min) is controlled with the help of a gearbox and by changing the voltage applied to the motor from the electric power supply. Electrolysis conditions have to provide a part of the electrode of a constant length to be submersed into the electrolyte. After electrolysis, a tow of silver threads is washed to remove any adhering salt. Proper combinations of current and cathode lifting rate allow producing fibers with a length up to 30 cm and various structures (Fig. 25.1).

PRODUCTION OF GOLD POWDERS

Gold-bonding conductors, owing to their high reliability and conductivity, find a variety of applications in electronics and communication systems. The bonding behavior of gold as well as its electrical conductivity make it desirable in hybrid microcircuitry. A special reliability of gold is connected with its superior resistance to oxidation. The primary cost of using gold is rather high, but lost operating time and repair expenses as a result of the deterioration of electrical properties of nongold elements due to the formation of oxide or sulfide films on contact surfaces are much higher [9].

Production of particles 3–180 μm in size is realized by gold vaporization in an inert gas atmosphere and condensation in ethanol. Ultrafine gold powder particles are produced from the solution based on organic complexes by thermolysis of its aerosols (drop diameter is of 1–5 μm) in an inert medium at temperatures higher than 770 K.

Many processes for manufacturing gold powder use an aqueous solution of H₂AuCl₄. This gold chloride solution is produced by dissolving gold grain or shot in aqua regia followed by heating to decompose and remove the nitric acid. A variety of reducing agents can be used to reduce gold to gold powder, including other metal powders (zinc, magnesium, aluminum,

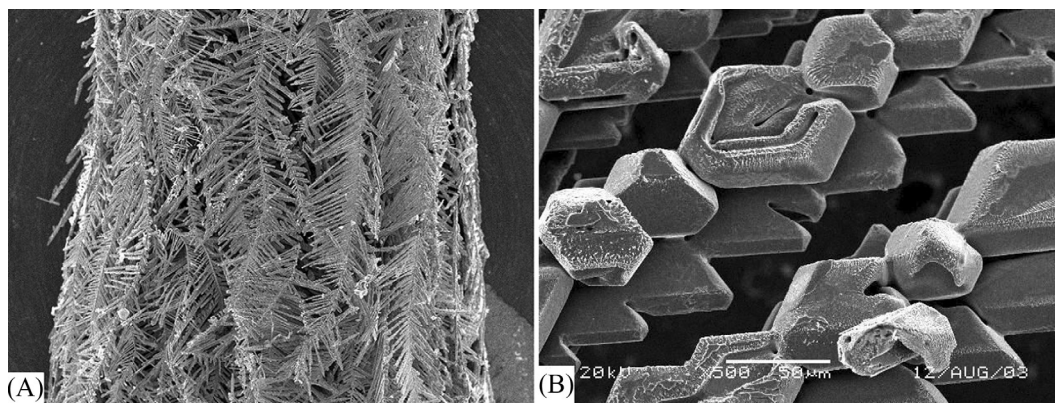
TABLE 25.3 Electrolysis Conditions and Process Parameters of Noble Metal Powder Production

Method	Electrolyte Composition (Content of Components, kg/m ³)	Current Density, kA/m ²	Temperature, K	Electrolysis Parameters
Palladium				
WS	10 of Pd (pH 9)	0.5–7.0	293	Electrolyte: Pd(NO ₃) ₂ + NH ₄ OH; current efficiency is of 40%
Platinum				
WS	3–5 of Pd(NH ₃) ₂ (NO ₂) ₂ ; 100 of K ₄ P ₂ O ₇ · 3H ₂ O; 50 of NH ₄ OH; (pH 10.5)	1.0	333	Current efficiency is of 19%–20%; deposit is black powder; apparent density is of 5 g/cm ³
Silver				
WS	32–48 of AgNO ₃ ; 3–5 of HNO ₃ ; 50–100 of NaNO ₃	$i_k = 1.2$; $i_a = 0.3$ –0.4	313–323	Current efficiency is of 85%–96%; voltage is 2 V
WS	15–20 of Ag; 10–15 of HNO ₃	1.5	313–323	Time for deposit growth is 2 min, interelectrode space is 50 mm, voltage is 2–2.8 V; electric power consumption is 0.6 kWh/kg
WS [3]	20 of AgNO ₃ ; 10 of NaNO ₃ ; up to pH 1 of HNO ₃	$i_k = 2.0$; $i_a = 0.5$	298	Current efficiency is 97%–100%; particle size is 2–3 μm
WS	20–300 of H ₂ SiF ₆	0.3–5.0	293–323	Silver anodes, stainless steel cathodes; 40%–80% of particles with size 60 μm, apparent density is 1.1–2.5 g/cm ³
WS [4]	15–20 of Ag; 100–200 of NH ₂ SO ₃ H; (pH 0.5–1.5)	0.6–0.8	291–298	Current efficiency is 90%–100%; particle size is 50–100 μm; mass use ratio is 60%–70%
EDC	50 of AgNO ₃ ; 0.5% of oleic acid; (pH 1–2)	1.2	293	Graphite disc is used for cathode; rotation is 60 rpm; precipitation time is 1 h; voltage is of 10 V; current efficiency is 99.8%; powder surface is 15–18 m ² /g
Palladium-Silver				
WS [5]	10 of Pd + Ag (pH 9)	0.5–7.0	293	Current efficiency (42%) depends on total metal concentration. Particle size is of 1–20 μm $(\text{Pd/Ag})_{\text{powder}} = (\text{Pd/Ag})_{\text{solution}}$
Palladium-Platinum				
WS [6]	10–100 of Pd + Pt; (pH 2.5–2.7)	0.5–4.0	293	Particles size is 1–10 μm; alloys are solid solutions; salts are chlorides; $(\text{Pt/Pd})_{\text{powder}} = (\text{Pt/Pd})_{\text{solution}}$
Platinum-Cobalt				
WS [7]	3–5 of Pt(NH ₃) ₂ (NO ₂) ₂ ; 5–8 of CoSO ₄ · 7H ₂ O; 100 of K ₄ P ₂ O ₇ · 3H ₂ O; 50 of NH ₄ OH; (pH 10.5)	1.0–2.0	333	Black powder, current efficiency is 20%; $(\text{Pt} + \text{Co})_{\text{sum}} = 5$ –10 kg/m ³ ; at $(\text{Pt/Co})_{\text{solution}} = 2/1$ $(\text{Pt/Co})_{\text{alloy}} = 1:1$ (at); residue is absent under $i \leq 100$ –200 A/m ²
Silver-Nickel				
WS	30 of Ag + Ni; 5–10 of Ag; 20–25 of Ni; (pH 4–6)	2.5–3.0	291–298	Sulfamic electrolyte; in solution Ni:Ag = 5:1; current efficiency is 40%; anodes–Ag; Ni = 1:4, the higher temperature, the richer the silver alloy

TABLE 25.3 Electrolysis Conditions and Process Parameters of Noble Metal Powder Production—cont'd

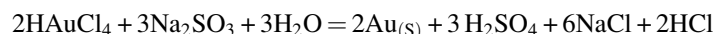
Method	Electrolyte Composition (Content of Components, kg/m ³)	Current Density, kA/m ²	Temperature, K	Electrolysis Parameters
Palladium-Nickel				
WS	4–14 of Pd+Ni; 20 of NH ₄ Cl; (pH 9.5)	10–15	293	[Pd/Ni]=2/1; solid solutions; (Pd/ Ni) _{powder} =(Pd/Ni) _{solution} ; particles size is 85–107 μm; current efficiency is 65%
Palladium-Copper				
WS	5–13.5 of Pd+Cu; 30 of NaCl; 60 of NH ₄ Cl; (pH 0.5)	3–4	293	(Pd/Cu) _{powder} =(Pd/Cu) _{solution} ; particle size is 90–220 μm; current efficiency is 65%; alloys are hard solutions

WS: water solution electrolysis; EDC—electrolysis from double-layer cell (see Paragraph “Reduction from solution through film of surfactants” of Chapter 8).

**FIG. 25.1** Silver fibrous deposits (current (i_a), A; velocity of electrode lifting (v), cm/min): (A) $i_a = 1$ A, $v = 5$ cm/min; (B) $i_a = 5$ A, $v = 3$ cm/min.

and iron), sulfur compounds (sulfur dioxide, sodium sulfite, potassium sulfite, and ferrous sulfate), hydrazines, aldehydes, oxalates, hydrogen peroxide, and others.

Potassium sulfite or sodium sulfite can be directly added to an aqueous gold chloride solution to produce spherical gold particles [10]:



Gold powder can also be produced by mechanical comminution. Sheets of gold are reduced by rolling to produce a foil between 10 and 20 μm thick, followed by beating to produce a gold foil of 0.2–0.5 μm thick. These foils are subjected to mechanical comminution, ensuring an ultrafine gold powder production. Larger gold powders and gold flake can be manufactured by ball milling [11].

PRODUCTION OF PLATINUM-GROUP POWDERS

Powders of Platinum Metals

The chemical composition is shown in Table 25.4. Powders with metal content not lower than 99.9% are produced according to special specifications. Each lot of powder corresponds to one brand of metal; foreign mechanical inclusions are not allowed.

Indium powder. Particle size of the powder is not greater than 0.8 mm. The powder is used for the manufacture of alloys, special coverings, and other products. The powder is supplied in lots of 250 kg.

TABLE 25.4 Chemical Composition of Refined Powders of Platinum Metals

Chemical Composition, wt%												
Metal, Brand ^a	Basic Metal Content, Min	Platinum Metals, Total	Impurities, Max									
			Pb	Fe	Si	Ba	Mg	Au	Al	Volatile Matters	Sn	Total
<i>Iridium</i>												
IA-1	99.95	0.02	0.005	0.01	0.002	0.002	0.001	0.002	0.005	0.01	—	0.05 ^b
IA-2	99.90	0.045	0.01	0.02	0.005	0.005	0.003	0.002	0.005	0.01	—	0.10
<i>Palladium</i>												
Pd AP-1	99.95	0.025	0.005	0.01	0.005	—	—	0.005	0.005	—	0.001	0.05
Pd AP-2	99.90	0.050	0.005	0.02	0.005	—	—	0.010	0.005	—	0.005	0.10
<i>Platinum</i>												
PL AP-0	99.98	0.015	—	0.003	0.002	—	—	—	—	—	—	0.02 ^c
PI AP-1 ^d	99.95	0.025	0.005	0.010	0.005	—	—	0.005	0.005	—	0.001	0.05
PI AP-2 ^e	99.90	0.050	0.005	0.010	0.005	—	—	0.005	0.005	—	0.005	0.10
<i>Rhodium</i>												
Rh-A1	99.95	0.02	0.005	0.01	0.005	0.005	0.001	0.002	—	0.01	—	0.05 ^f
Rh-A2	99.90	0.03	0.005	0.02	0.005	0.005	0.003	0.002	0.002	0.02	—	0.10
<i>Osmium</i>												
OsA-0	99.97	—	—	0.01	—	—	—	0.002	—	—	—	0.03 ^g
Os A-1	99.95	—	—	0.01	—	—	—	0.002	—	—	—	0.05
OsA-2	99.90	—	—	0.03	—	—	—	0.002	—	—	—	0.10
<i>Ruthenium</i>												
Ru A-0	99.97	—	0.003	0.003	0.003	0.002	—	0.002	0.002	—	—	0.03 ^h
Ru A-1	99.95	—	0.005	0.01	0.005	0.005	—	0.002	0.005	—	—	0.05
Ru A-2	99.90	—	0.01	0.02	0.01	0.005	—	0.002	0.005	—	—	0.10

^aIn CIS classification.^bIncluding Ag, Ni, and Cu.^cIncluding Ag, Mg, Cu, Ni, and volatile matters.^dIncluding 0.001 of Sb.^eIncluding 0.005 of Sb.^fIncluding Ag, Cu, Ni, Ti, and Al.^gIncluding Pt, Pd, Rh, Ir, Ag, Cu, and Ni.^hIncluding Pt, Pd, Rh, Ir, Ag, Cu, Ni, Mg, O, and the volatile matter Si, Al, Mg, Ba, Na including.

Palladium powder. Particle size of powder is not greater than 1.6 μm. The powder is supplied in lots of not more than 350 kg. It is used for the manufacture of contacts, alloys, chemicals, catalysts, special coatings, and other products.

Platinum powder. Particle size of the powder is less than 1.6 μm; a content of the size fraction larger than 1.6 μm up to 2% is allowed. The mass of a powder lot is not larger than 350 kg. The powder is used for the manufacture of special units, alloys, coatings, semifinished items, and chemicals.

Rhodium powder. Particle size of powder is not greater than 0.8 μm; it is packed into plastic cans (net weight 25–5000 g) or in welded glass capsules (net weight 0.1–25 g). The mass of the powder lot is not greater than 350 kg. It is used for the manufacture of alloys, products, and chemicals.

Osmium powder. Particle size of the powder is less than 1.0 μm; inclusions of the size fraction larger than 1.0 μm (up to 2%) are allowed. The powder is packed into glass containers (25–5000 g) and capsules (0.1–25 g). It is used for the manufacture of alloys, semifinished items, special products, coatings, and chemicals.

Ruthenium powder. Particle size of powder is below 1.0 μm; inclusions of the size fraction larger than 1.0 μm (up to 2%) are allowed. The powder is packed into plastic containers (25–5000 g) and glass capsules (0.1–25 g). It is used for the manufacture of alloys, semifinished items, special products, coatings, and chemicals.

Chemical Reduction

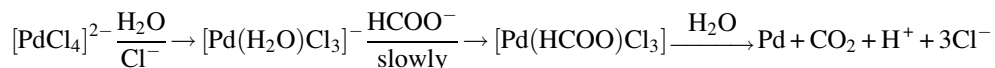
The precipitation of platinum oxide at the interaction of H_2PtX_6 (where X is Cl^- , Br^- , I^-) with NaNO_3 and the processing of the oxide in a hydrogen atmosphere is the oldest method of platinum niello production. Dispersivity is controlled at the stage of platinum oxide precipitation by variations of the salt nature and the inclusion of halides of alkali-earth metals (the latter are typical additions).

The process of reducing $[\text{PtCl}_6]^{2-}$ ion by sodium formate includes a number of consecutive stages: $\text{Pt(IV)} \rightarrow \text{Pt(II)} \rightarrow \text{Pt}$; the higher the (Cl^-) ion concentration, the slower the process rate. The activation energy for the interval of 368–333 K is 105 kJ/mole.

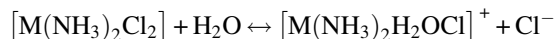
The fractional conversion of the metal to powder achieves 99.9% in processing of chloro-amine compounds of platinum and palladium in a sodium hydroxide solution by reducing agents (sodium formate, aluminium, hydrazine salts).

Platinum niello is produced by the reaction of ammonium hexachloroplatinate with a 2.5-M solution of formic acid or sodium formate at 375 ± 5 K; a pH of 6.0–6.5 is supported with the help of NaOH (or NH_3).

Metallic palladium powder is formed (without reducer) in the process of boiling dichloridiamminpalladium salt in a saturated alkali solution. If this salt is treated slowly (3 h at 375 ± 5 K) in a sodium hydroxide solution and sodium formate (0%–20% of stoichiometric consumption) is used as the reducer, we can get palladium niello. An increase of sodium formate consumption from 10% to 80% of the stoichiometrically necessary level results in a decrease of palladium niello specific surface from 32 to $3 \text{ m}^2/\text{g}$. Palladium precipitation from chloride solutions by sodium formate is performed in accordance with the following scheme:



The greatest influence on the increase of the process rate is exerted by an increase of temperature, sodium formate concentration, and seed consumption. The process rate decreases at high pH and especially if chloride ions are introduced. It is explained by the enhanced resistance of metal chloride components to the activation:



With the appearance of the metallic niello, the process becomes autocatalytic.

Palladium powder is produced by processing its ammonium complexes with a mixture of formic acid and with its ammonium salt prepared in the following manner: $(\text{CH}_2\text{O}_2:\text{NH}_4\text{OH} \geq 1.0:0.3)$ at 350–360 K.

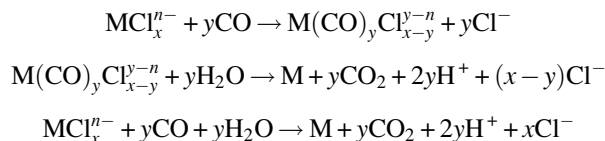
Cladding of powders of various metal oxides by platinum or palladium is achieved by the thermolysis method.

Powders of platinum, palladium, ruthenium, and iridium are precipitated from an acid-salt solution by cementation with zinc, copper, magnesium, aluminum, or cadmium. The powder surface area is $0.1\text{--}1.0 \text{ m}^2/\text{g}$. A possible formation of inter-metallics with the precipitator used reduces the powder quality.

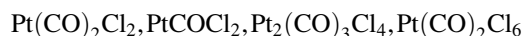
Fine powders of platinum group metals are precipitated from aqueous or nonaqueous solutions of salts by radiation by an electron beam in doses of 10–30 mrad.

The Carbonyl Process

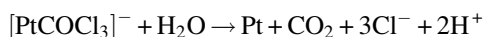
Powders of platinum group metals can be precipitated by the use of carbon monoxide:



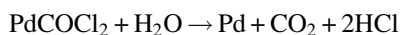
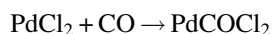
There are several platinum carbonyl compounds:



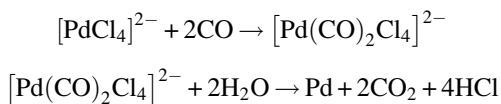
High carbonylchlorides are hydrolyzed to monocarbonylchloride; the latter is decomposed by interaction with water forming the platinum powder:



Platinum metals may exist in chloride solutions in different ion forms according to the concentration of chloride ions.



Palladium complexes with maximal coordinate number 4 are formed in concentrated chloride solutions. In accordance with the above mentioned mechanism:



This process includes a number of subsequent transformations with complex ions of different structures as the participants. The reduction is performed from the solutions (100–120 g/L of Pd; 1–2 M of HCl) or the H_2PdCl_4 salt solution, which is neutralized by alkali to a pH of 6–8 and precipitated at 340–345 K.

The rate of palladium precipitation decreases in the presence of organic compounds (acetone, ethanol, butane). When the organic phase concentration is increased, the fine fractions output is increased also, and practically monodispersed powder can be produced. Palladium powder with a particle size of 1.2–1.7 μm was reduced in the system palladium salt solution + about 10% of metal; the particle shape is nonequiaxed, and the structure is porous.

The properties of palladium powders used for the manufacture of thin-walled ceramic condensers are presented in Table 25.5.

Characteristic properties of palladium powders are homogeneity of phase and chemical composition, high particle size dispersity (0.3–3.5 μm), and insignificant pyrophorosity, though of large specific surface (3.5–16.0 m^2/g). Apparent density

TABLE 25.5 Properties of Palladium Powders Used for the Manufacture of Thin-Walled Ceramic Condensers

Property	Palladium	Alloy
Palladium content (min %)	97	96
Chlorine-ion content (%)	0.01	0.04
Tap density (g/cm^3)	0.8–1.0	3.0–4.0
Specific surface (m^2/g)	10–16	3.5–6.0
Average particle size (μm)	0.3–0.6	2.0–3.5
Beginning of oxidation (K)	470–490	820–870
Phase composition	Pd	Pd + (5–10)% of PdO
Degree of transformation to oxide (%)	100	15–20

risks with the increase of the concentration of palladium salt, hydrochloric acid in the solution and the temperature and amounts to 0.8–2.8 g/cm³. The specific surface is most significantly influenced by the content of palladium and acid in the solution, and the temperature as well. To precipitate 1 kg of palladium, 262 g of carbon monoxide with an 80% level of usage is required.

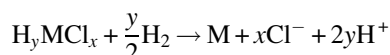
Osmium powder is produced from platinum containing 0.6%–1.0% of Os in two steps, according to the scheme OsO₄ oxide sublimation–electrolysis. Melting is done in an oxidizing atmosphere, and sublimates are collected in a 20% NaOH solution. Osmium-bearing powder (50%–70% of Os) is produced after the first stage by electrochemical precipitation. It is retreated in the same way; the final cathode residue is washed free from impurities, dried, and calcinated in a hydrogen atmosphere.

Autoclave Precipitation

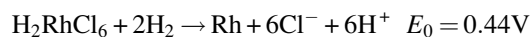
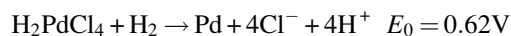
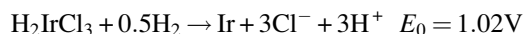
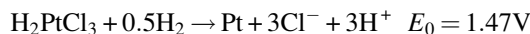
Complex compounds are stable forms of noble metal ions in aqueous solutions. The thermodynamic probability of their autoclave reduction by hydrogen is sufficient significant. The development of the process of metal powder precipitation processes is determined by the kinetic parameters and courses at less rigid parameters than the precipitation of copper, nickel, and cobalt powders.

The precipitation of gold from its thiosulfate, chloride, and bromide complexes with 3+ charge proceeds at 330–340 K.

V.V. Ipatyev (1926) had first revealed the possibility of platinum precipitation from the chloride complex. He carried out this process at 293–303 K, $p_{H_2} = 2.5\text{--}4.0$ MPa within 18–40 h. A positive influence of a seed was established, and the precipitation rate increased in the series Pt, Ir, Rh, and Pd. In general, this process can be described by the following reactions:



Similar reactions for actual platinum metals are presented below:



Palladium and rhodium were reduced from the solutions containing 15–25 ppm (parts per million) of metal and 1 M of HCl at a hydrogen pressure of 0.1 MPa at 293–313 K, and platinum and ruthenium were reduced at 343–353 K. The effect of increasing p_{H_2} to 0.35 MPa is established only for platinum and ruthenium.

In processing the acetate sulfate solutions, the reduction of Rh (III), Pt (IV), and Ru (III) to metal proceeded rather quickly, and the process rate did not depend on the anion concentration and the ionic force of the solution. Cl[−], I[−], NH₃ ligands at their content higher than 3N decelerate platinum metal precipitation, and the process stops in the presence of CN[−] ($p_{H_2} = 0.01$ MPa, 353 K). There is an induction period in kinetic curves: the process rate is described by the equation of the first order of $-d[M]/d\tau = k[M^{n+}] \cdot [H_2]$ type. Experimental rate constants k (s^{−1}) of reduction are as follows:

$$\begin{aligned} k : \text{for RuCl}_6^{3-} &= 1.0; \quad \text{for RhCl}_6^{3-} = 0.18; \\ \text{for PdCl}_4^{2-} &= 0.33; \quad \text{for PtCl}_4^{2-} = 5 \times 10^{-4} \end{aligned}$$

The precipitation of palladium from the solution bearing 2 g/L of Pd as H₂PdCl₄ requires the following: 303 K, $p_{H_2} = 0.3$ MPa; $\tau = 1$ h; the process is not developed in the presence of Cu (II) or Fe (III) ions.

Iridium can be reduced only to Ir (III) during the processing of chloride solutions bearing Ir (VI) ions at 303 K. Iridium metal is precipitated from the solution (64 mg/dm³, 20 g/L of HCl) only at 413 K, $p_{H_2} = 0.3$ MPa within 1 h. Platinum metals are precipitated at 353–420 K within 0.5–1.0 h with sufficient stirring. Selective precipitation can be carried out depending on temperature and hydrochloric acid concentration.

Homogeneously sized silver powder with a bulk weight of 0.3–3.0 g/cm³ is produced by processing silver oxide pulp bearing 10–200 g/L of solid, $T = 330\text{--}390$ K, and $p_{H_2} = 0.5\text{--}2.5$ MPa within 10–40 min. Ammonia or silver salt is added to the pulp to increase the bulk weight, and alkaline-earth metal sulfides or silver powders are added to decrease it. The powder produced is used for the manufacture of products, compositions, and catalysts.

Noble metal powders can be precipitated by carbon monoxide in the interval of $\text{pH} \leq 7.0$, at temperature 293–473 K, and $p_{\text{CO}} \geq 0.1 \text{ MPa}$.

PRODUCTION OF NOBLE METAL ALLOYS POWDERS

The key condition of alloy production is phase composition uniformity achieved by alignment of the precipitation rate of components and their joint crystallization.

The most popular methods are chemical, contact displacement, and electrochemical precipitation.

Silver-palladium powders used, in particular, as metallized paste fillers in capacitor production have to meet the following requirements: 70% of Ag and 30% of Pd, surface of not lower than $1.5\text{--}2.3 \text{ m}^2/\text{g}$, apparent density of $0.7\text{--}1.0 \text{ g/cm}^3$, content of the fraction with the particles of size $\geq 90 \mu\text{m}$ not higher than 15%, and a homogeneous phase composition.

For their production, the basic raw material is dissolved in nitric acid, then the solution (80–100 g/L of Ag and Pd) is neutralized by alkali to $\text{pH} 4.5\text{--}5.0$, and the powder is precipitated by sodium formate to the residual content $\Sigma(\text{Pd}, \text{Ag}) < 1 \text{ mg/L}$ at 340–350 K. The reducer consumption is 1.5–2.0-times the stoichiometric composition. The produced powder is annealed in a reducing atmosphere and screened. Due to a developed surface ($>5\text{--}7 \text{ m}^2/\text{g}$) and the presence of residual nitrates, the oxygen content in the powder can rise to 1.5%–2.0%, which is intolerable.

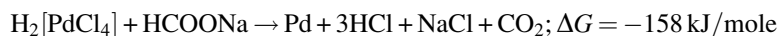
Powders with an average particle size of $0.6 \mu\text{m}$ were produced using hydrazine as a reducer (320–330 K, six times the consumption of stoichiometric quantity). According to another method, the silver-palladium composition is produced by stirring nitrate solutions (17 g/L of Ag, 3 g/L of Pd) with a reagent-precipitator (hydrazine + surfactants) at $326 \pm 5 \text{ K}$. The surface of the produced powder is $2\text{--}7 \text{ m}^2/\text{g}$.

The Pd-Cu alloy (55% of Pd) is produced from the solution of chlorides of these metals by reduction with hydrazine-sulfate. Two-three component alloys are produced by precipitation from hydroxy-ammonia and chloride ammonia complexes (the latter to provide the phase composition uniformity) of the $(\text{NH}_4)_2\text{MCl}_4$ type where M is Pt (IV), Pd (IV), Os (IV), and Ir (IV) with further treatment of the powder in hydrogen at 670 K.

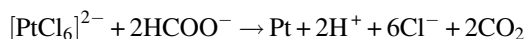
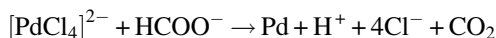
A local reduction, first of all of palladium, is possible when using a pulp of salts of metals, which are constituents of the produced alloy, and its precipitation by an organic reducer (most often sodium formate). Further, this results in a nonhomogeneous distribution of solid solution components in the volume of a unit particle and a formation of up to 2%–5% of individual metal inclusions.

Powders of Pt-Pd alloy are produced by joint precipitation by treatment of a solution of chloride complexes of these metals with a sodium formate solution:

at $\text{pH} < 3$

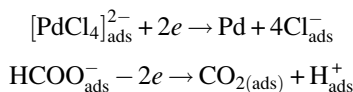


at $\text{pH} > 3$



The process includes: (1) induction, (2) precipitation and growth of nuclei, and (3) desorption and withdrawal into the solution of reaction products:

1. The induction stage is the diffusion of complex ions of the metal and the reducer to the seed surface.
2. The stage of precipitation and growth of nuclei is adsorption and an oxidation-reduction reaction:



accompanied by the growth of nuclei.

3. The stage of reaction product desorption and their diffusion into the solution.

In the case of multiple crystallization, the limiting stage is diffusion of Pt (Pd) (II) compounds to crystal surfaces. With the increase in pH , a part of aqua and hydroxo forms in the composition of the metal complex ions as well as the probability of niello precipitate formation.

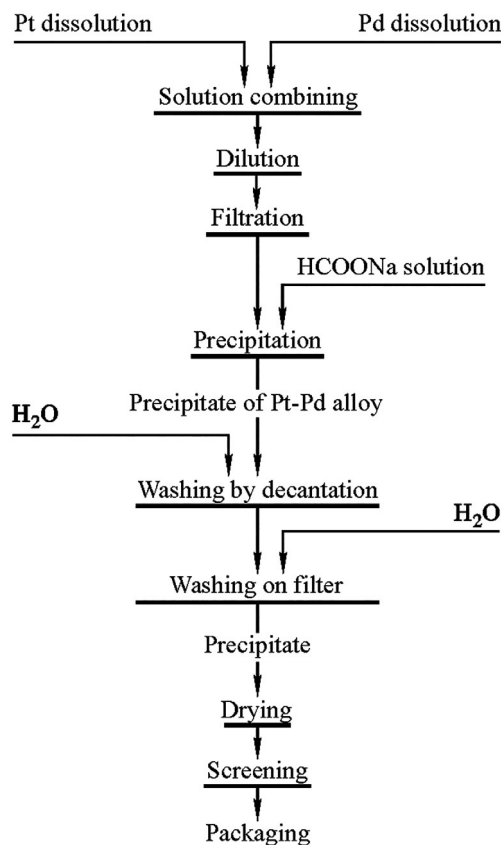


FIG. 25.2 Flow-sheet of Pt-Pd chemical precipitation from solutions.

The basic diagrams of alloy chemical precipitation are shown in Figs. 25.2 and 25.3. They differ according to the type of basic raw material: a slurry of metal salts or a solution. The proportion of components in the material that takes part in the precipitation provides further production of the alloy with a predetermined composition. It is adjusted in the starting material: for example, when producing the alloy PtPd-40p (in Russian classification) the proportion $[Pt]_0:[Pd]_0 = 1.4 \pm 0.1$ is kept in the solution. The precipitation is performed by a sodium formate solution ($\gamma = 1.25$) at 435 ± 2 K, $pH > 7$. The precipitate is filtered out and thoroughly washed for complete elimination of Cl (I) and Na (I) ions. The precipitate is dried for 4–6 h at 342 K and for 20–24 h at 393 K. It is then screened and packed.

The composition of the alloy of PtPd-40p brand (in CIS classification) is: $\Sigma(Pt, Pd) > 96.5$ wt% including 58–61 wt% of Pt; not more than 0.04 wt% of Cl; 0.03 wt% of Na; 0.5 wt% of moisture; and 0.05–1.0 wt% of volatile matter (O_2 , H_2 , CO_2 , CO).

This alloy is a single-phase alloy; its bulk density is of $1.4\text{--}1.9$ g/cm³. For example, per 1 kg of PtP-90 niello precipitate produced from metal, the consumption (in kg) is the following: 2.4 of HNO_3 ; 9.0 of HCl ; 2.4 of Na_2CO_3 ; 3.0 of $HCOONa$; 200 of H_2O ; and 100 kWh of electric energy. In particular, this alloy is, like platinum, suitable for the metallization of pottery surfaces bearing bismuth or titanium. In the processing of ammonium complexes of platinum and palladium (pH 12–13; 420–430 K; $L:S=2$; $\tau=0.5\text{--}1.0$ h), the latter are precipitated fully as powder.

The technique provides a quick and complete procedure with the minimum consumption of alkali and without an external reducer. The particles have a spherical form, their size is $0.3\text{--}0.4$ μm , and the specific surface of the powder is $2\text{--}4.5$ m²/g [11]. The size of the particles decreases when the temperature of the process increases, and the specific surface of the powder increases slightly.

Pt-Pd-Rh alloy is produced by thermolysis of a mixture of complex salts of $[M(NH_3)_4][NH_4(RhCl_6)]$, $M(NH_3)_4[RhH_2OCl_5]$ type where M is Pt (II), Pd (II). The products are solid solutions with compositions preset in the stage of mixture preparation; the powder particles have a spherical shape and the particle size is ≥ 10 μm .

A powder with apparent density of $0.7\text{--}1.9$ g/cm³ and a porous structure of particle conglomerates $30\text{--}40$ μm in size is produced by thermolysis of platinum and palladium glycine complexes.

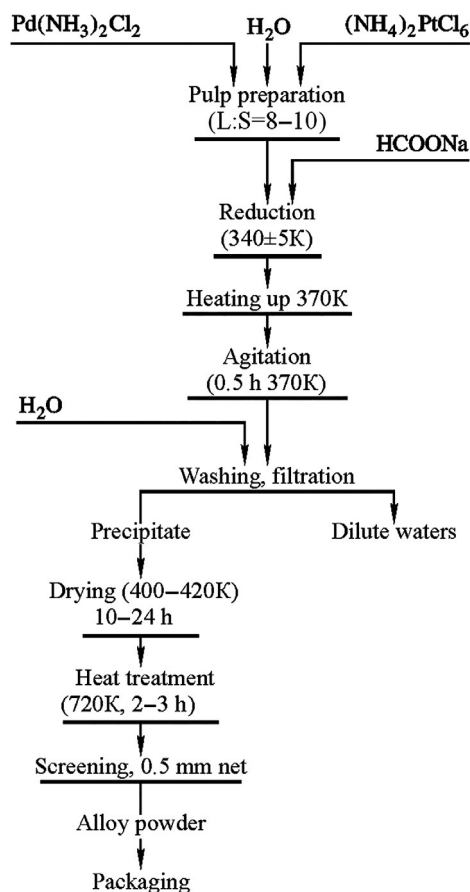


FIG. 25.3 Method of Pt-Pd chemical precipitation from pulp.

Powders of palladium-based compositions were produced by thermolysis of aminooxalate complexes of $[\text{Pd}(\text{NH}_3)_4][\text{M}(\text{C}_2\text{O}_4)_2]$ type where M is Zn, Cu, Ni, Co. An intermetallic was produced in the Pd-Zn system while, in other cases, the solutions were mixed with the metal in the ratio of 1:1. The powder particle size for all compositions is not larger than $1\mu\text{m}$.

An isomorphous solution of platinum in palladium is precipitated by hydroformulation of a solution bearing platinum and palladium. The larger the palladium content, the greater the process rate. Platinum is precipitated sufficiently completely at the atomic proportion $\text{Pt/Pd} = (1-2)/1$. The procedure is performed at 343K , $[\text{HCl}]_0 = 0.75\text{M}$, $[\text{NaCl}]_0 = 1.25\text{M}$. Increasing the Cl (I) ion concentration decelerates the process, but increasing acidity accelerates it; the powder particle size is $1.8-2.7\mu\text{m}$. Temperature and duration are governing factors for Pt-Pd-Rh mixture precipitation: 26% of the metal was precipitated at 370K within 3 h while 97% was precipitated in 8 h.

PRODUCTION OF PLATINUM BLACK DEPOSIT (NIELLO)

Before platinization, the electrodes are carefully cleaned in a chrome mixture or by SiO_2 powder, then calcinated. Platinization is carried out in the solution containing 30kg/m^3 of H_2PtCl_6 and $0.2-2.3\text{kg/m}^3$ of $\text{Pb}(\text{CH}_3\text{COO})_2$ at a current density that provides a moderate gassing on the electrodes ($\approx 50\text{A/m}^2$) and at 295K within 15 min. In the case of simultaneous platinization of two electrodes (for example, for application in cells of electrical conductivity measurement), the current is commutated from time to time.

After platinization, the electrodes are put into a 10% solution of sulfuric acid and polarized cathodically within 5 min with an auxiliary platinum electrode as the anode. The current strength in this case is adjusted also by gassing intensity. The adsorbed chlorine is reduced and eliminated more easily while washing as chloride ions. The electrodes are washed in running warm water and then in cold distilled water. The platinized electrodes have to be stored under a water layer as they lose wettability in air. Replatinization is performed in 1-2 min.

Synthesis of Palladium Nanopowders by Methods of Zone Electrochemistry

Palladium powder with a particle size of 5–15 nm is characterized by specific catalytic properties. Nanoparticles and their dendrite-shaped aggregates are produced from palladium chloride by means of cetyltrimethylammonia bromide (CTAB). The cylindrical part of the titanium rod is protected by plastic, and rectangular pulses of direct current and ultrasound are supplied alternatively to the electroactive disc part of the electrode. A predominant size of spherical particles (nuclei stripped from the electrode surface during ultrasound supply) is increased from 5 to 11–13 nm as the current density decreases from 130 to 80 A/m² in the solution with a palladium salt concentration of 10 mol/m³. A preferred concentration range of palladium salt is 5–50 mol/m³, and the interval between adjacent ultrasound pulses is 5–10 s. The CTAB content can vary within a range from 20 to 30 kg/m³. Dendrite aggregates from nanoparticle chains are formed with increasing electrolysis duration to 2.5 h [12].

Manufacturing methods for electrolytic platinum-group powders and electrolytic platinum-group alloy powders are shown in Table 25.3.

Producing of Noble Nanoparticles in Metal-Organic Frameworks Pores

Metal-organic frameworks or porous coordination polymers continue to attract much attention in new emerging areas, particularly those related to energy use and environmental protection. The ZIF-8 framework (Zn(mim)₂, mim is 2-methylimidazole) has become popular for use as a support of select metal nanoparticles (M-NPs) for heterogeneous catalysis [13–15].

Zhang et al. [16] reported that an integral part of a porous framework and uniformly distributed throughout the internal pore space, the high density of the exposed B–H bond in zeolite-like porous BIF-20 (BIF = boron imidazolate framework) showed an effectively produce nanoparticles within its confined pore space. Small noble-metal nanoparticles (silver and gold) are directly synthesized into its pores without the need for any external reducing agent or photochemical reactions, and the resulting Ag@BIF-20 (or Au@BIF-20) samples show high catalytic activities for the reduction of 4-nitrophenol. The weight percentage of the silver nanoparticles in BIF-20 is 3.01%.

Noble Metals Application in Implantable Medical Devices

Implantable medical devices (cardiac defibrillators, cochlear implants, or neuromuscular micro stimulators) have been used worldwide to restore body functions, improve the quality of life, and save lives. Implantable medical devices are implanted long term and contain electronic circuits and batteries. In addition, for a safe and sustainable implantation in the human body without the release of toxic products and device degradation by body fluids, a hermetic housing is required. Generally, conductive pads that are surrounded by a biocompatible insulating material [17] ensure the communication between an electronic system located in hermetic packaging and sensors on the human body. A hermetic electrical feedthrough is thus essential for a safe and functional active implantable biomedical device. It needs to be as small as possible for patient comfort and safety reasons.

To bond metal and ceramic materials, brazing is so far the most widely used technique when a mechanically reliable gas tight bonding is required. However, for implantable devices, a high content of expensive noble metals is often added into the brazing paste, such as gold. The commonly used ceramic/metal feedthrough is made of pure platinum pins surrounded by alumina [17]. This kind of complex multimaterial component fabrication by a conventional process needs complex and expensive steps due to the dimensional accuracy required by each part and the brazing process.

Powder injection molding (PIM) can open new opportunities to manufacture hermetic feedthroughs between metal electrodes, ceramic insulators, and metal rings for packaging welding. The process allows manufacturing complex multimaterial and gas tight components in a reduced number of producing steps. Moreover, it allows, with an appropriate choice of materials and treatment conditions, bonding ceramic and metal without brazing. This work proposes to explore the PIM potential to carry out a hermetic feedthrough without brazing.

Two feedthroughs for medical implantable devices were carried out by PIM [18]. The metal-ceramic interfaces formed, Pt/Al₂O₃ and Al₂O₃/Ti-6Al-4V, were investigated. Pure platinum wires with 0.33 mm diameter are used. Wires were cut to obtain 25 mm length pins. Then, the pins were cleaned in an ethanol/acetone bath. An alumina feedstock (a mixture of ceramic powder and binder) was supplied by Inmatec GmbH, Germany. CT1200 alumina in the feedstock with a 99.8% purity was used.

The leakage measurements performed on a circular feedthrough show that, for parts sintered under argon or vacuum, the rate of leakage is lower than 1·10^{−9} mbar. After alumina sintering, the leakage rate measurement on the Pt pin/alumina

interface was lower than $1 \cdot 10^{-9}$ mbar. To simulate the second thermal treatment, pin/alumina parts were heated at 1523 K under vacuum or argon. The leakage rate was also found to be below $1 \cdot 10^{-9}$ mbar. The same gas tightness measurement was also performed on parts used for tensile tests with the same results. All results show that the bonding between the platinum pin and alumina is strong enough to obtain a hermetic interface, fulfilling the feedthrough requirements.

REFERENCES

- [1] Butts A, Cox CD, editors. Silver—economics, metallurgy, and use. Malabar: Krieger Publishing; 1975.
- [2] Glicksman HD. Production of precious metal powders: silver, gold, palladium, and platinum. In: ASM Handbook, vol. 7. ASM International Publishers; 1998. p. 182–7.
- [3] Baimakov YV, Zhurin AI. Electrolysis in hydrometallurgy. Moscow: Metallurgiya Publisher; 1977 [in Russian].
- [4] Tomiltseva AK, Vyacheslavov PM, Burkat GK, Tsoi ZN. Silver powders electroprecipitation from sulphamine electrolyte. Soviet J Appl Chem 1978;51(9):2094–7 [in Russian].
- [5] Yurev BP, Shkuryakova SP. Electrolytic preparation of fine Pd-Ag powders of any given chemical composition. Soviet Powder Metall Metal Ceram 1973;12(1):443–6 [in Russian].
- [6] Shkuryakova SP, Yurev N. Production of fine platinum-palladium alloy powders. Soviet Powder Metall Metal Ceram 1974;13(12):953–5 [in Russian].
- [7] Zayats AI, Perekhrest NA. Dispersed Pt-Co alloys precipitation. Soviet J Appl Chem 1972;45(10):2248–50 [in Russian].
- [8] Ivanovskiy LE, Rozanov IG, Zotin IV, Khramov AP. Production of dendrite metals deposits by ionic melt electrolysis and their application. Liquid Melts 1997;2:51–9.
- [9] Davies RR. New development in electronics and communications. In: RO MG, Bradley AG, editors. Precious Metals. Pergamon Publ.; 1981.
- [10] Short, O.A., U.S. Patent 3,771,996, 1973.
- [11] Collier N. Advanced in gold powder technology. Gold Bull 1977;10(3):62–6.
- [12] Qiu XF, Xu JZ, Zhu JM, Zhu JJ, Xu S, Chen HY. Controllable synthesis of palladium nanoparticles via a simple soundoelectrochemical method. J Mater Res 2003;18(6):1399–404.
- [13] Moon HR, Kim JH, Suh MP. Redox-active porous metal-organic framework producing silver nanoparticles from Ag^+ ions at room temperature. Angew Chem Int Ed 2005;44:1261–5.
- [14] Suh MP, Moon HR, Lee EY, Jang SY. A redox-active two-dimensional coordination polymer: preparation of silver and gold nanoparticles and crystal dynamics on guest removal. J Am Chem Soc 2006;128:4710–8.
- [15] Zlotea C, et al. Pd nanoparticles embedded into a metal-organic framework: synthesis, structural characteristics, and hydrogen sorption properties. J Am Chem Soc 2010;132:2991–7.
- [16] Zhang HX, Liu M, Bu X, Zhang J. Zeolitic BIF crystal directly producing noble-metal nanoparticles in its pores for catalysis. Sci Rep 2014;4(3923) <https://doi.org/10.1038/srep03923>, <http://www.nature.com/articles/srep03923> (Accessed 31 December 2017).
- [17] Jiang G, Zhou DD. Technology advances and challenges in hermetic packaging for implantable medical devices. New York: Springer; 2010.
- [18] Vincenti D, Delette G, Laucourne R. A Gas tight platinum-alumina-TA6V feedthrough for implantable medical devices without brazing by powder metal injection. In: compiled by European Powder Metallurgy Association, (Bellstone Shrewsbury, UK) Proceedings of PM 2017 European Congress, Milan (Italy); 2016: p. 1–6.